

INFOTAINMENT HEAD UNIT

High-Level Design Document

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Classification: Synthetic Demonstration Data

1. System Overview

The Infotainment Head Unit coordinates its functions through typed AUTOSAR interfaces and ports. Software components exchange signals through provider and consumer ports.

2. Software Architecture

The primary software components are MediaManager, AudioControl, DisplayControl, BluetoothManager, and VolumeControl.

3. Software Components

Component	Role	Primary Responsibility
MediaManager	Coordinator	Controls playback and source selection.
AudioControl	Consumer	Drives the amplifier and equaliser.
DisplayControl	Consumer	Renders the user interface.
BluetoothManager	Provider	Handles phone pairing and streaming.
VolumeControl	Provider	Reads steering-wheel volume buttons.

4. Interfaces

Interface	Provider	Consumer	Signals
IVolumeLevel	VolumeControl	AudioControl	VolumeLevel
IAudioStream	MediaManager	AudioControl	StreamState
IPhoneConnection	BluetoothManager	MediaManager	PhoneStatus
IDisplayContent	MediaManager	DisplayControl	TrackTitle

5. Ports

Component	Port	Direction	Interface
VolumeControl	VolumeLevel_Out	P-PORT	IVolumeLevel
AudioControl	VolumeLevel_In	R-PORT	IVolumeLevel
MediaManager	AudioStream_Out	P-PORT	IAudioStream
AudioControl	AudioStream_In	R-PORT	IAudioStream
BluetoothManager	PhoneConnection_Out	P-PORT	IPhoneConnection
MediaManager	PhoneConnection_In	R-PORT	IPhoneConnection
MediaManager	DisplayContent_Out	P-PORT	IDisplayContent
DisplayControl	DisplayContent_In	R-PORT	IDisplayContent

6. Signals

Signal	Data Type	Source	Destination
VolumeLevel	uint8	VolumeControl	AudioControl
StreamState	uint8	MediaManager	AudioControl
PhoneStatus	boolean	BluetoothManager	MediaManager
TrackTitle	uint32	MediaManager	DisplayControl

7. Dependencies

Source	Relationship	Target	Reason
VolumeControl	PROVIDES_INTERFACE	IVolumeLevel	Publishes volume level.
AudioControl	REQUIRES_INTERFACE	IVolumeLevel	Consumes volume level.
MediaManager	PROVIDES_INTERFACE	IAudioStream	Publishes stream state.
AudioControl	REQUIRES_INTERFACE	IAudioStream	Consumes stream state.
BluetoothManager	PROVIDES_INTERFACE	IPhoneConnection	Publishes phone status.
MediaManager	REQUIRES_INTERFACE	IPhoneConnection	Consumes phone status.
MediaManager	PROVIDES_INTERFACE	IDisplayContent	Publishes track title.
DisplayControl	REQUIRES_INTERFACE	IDisplayContent	Consumes track title.

8. Functional Flows

Flow F-001: Volume Level Distribution

VolumeControl publishes VolumeLevel through IVolumeLevel. AudioControl receives the signal through VolumeLevel_In and updates its internal state model.

Flow F-002: Stream State Distribution

MediaManager publishes StreamState through IAudioStream. AudioControl receives the signal through AudioStream_In and updates its internal state model.

Flow F-003: Phone Status Distribution

BluetoothManager publishes PhoneStatus through IPhoneConnection. MediaManager receives the signal through PhoneConnection_In and updates its internal state model.

Flow F-004: Track Title Distribution

MediaManager publishes TrackTitle through IDisplayContent. DisplayControl receives the signal through DisplayContent_In and updates its internal state model.

9. Integration Constraints

All provider and consumer ports shall reference compatible interfaces. Signal names shall remain consistent between interface definitions and functional flows. Architecture dependencies must be traceable to an approved source section.

10. Engineering Traceability

This synthetic HLD supports extraction, retrieval, validation and revision comparison experiments.